

Dimensional regularisation of the UV-projected row

$\overline{\text{MS}}$ done where it is defined — in momentum space

the $1/r^2$ term Fourier-transformed to the one-loop transverse bubble, then subtracted

0. Conventions

$D = 4 - 2\epsilon$, transverse plane $d \equiv d_\perp = 2 - 2\epsilon$. $\overline{\text{MS}}$ is a *momentum-space* prescription: the loop measure is $\mu^{2\epsilon} \int \frac{d^d \ell}{(2\pi)^d}$ and one subtracts

$$\frac{1}{\hat{\epsilon}} \equiv \frac{1}{\epsilon} - \gamma_E + \log 4\pi \quad \left(\text{equivalently, keep } \frac{1}{\epsilon} \text{ and write } \bar{\mu}^2 = 4\pi\mu^2 e^{-\gamma_E} \right).$$

Kinematics: $r = z - x$, $s = y - x$, $P = x' - w'$, $q = \bar{\xi}k$, $D = (r - s)^2 + M^2$, $M^2 = \frac{\xi}{\bar{\xi}}s^2$, $D_0 = D|_{r=0} = s^2/\bar{\xi}$,
 $C_{\text{UV}}(\xi) = \xi\bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} = \frac{P_{qq}}{2N_c}$, $\ell_k = \log \frac{k^+}{\Lambda}$.

Step 0. C_{UV} carries no $O(\epsilon)$. The instantaneous δ^{im} vertex supplies an explicit $d_\perp$ and the angular average supplies $1/d_\perp$; they cancel. Nothing multiplying the pole is itself $O(\epsilon)$, so no extra finite term is generated by $\epsilon \times \frac{1}{\epsilon}$.

1. What has to be regulated

$C_{\text{UV}}(\xi)$ and $(s \cdot P)/P^2$ do not depend on r , so the whole transverse integration is one object:

$$\mathcal{G} = \int d^2 r \frac{e^{iq \cdot r}}{r^2 D}.$$

2. Split the divergence off exactly

$$\frac{1}{r^2 D} = \frac{1}{r^2 D_0} + \frac{1}{r^2} \left(\frac{1}{D} - \frac{1}{D_0} \right), \quad \frac{1}{D} - \frac{1}{D_0} = \frac{2r \cdot s - r^2}{D D_0},$$

the second bracket vanishing linearly at $r = 0$. So only the first term needs a regulator:

$$\mathcal{G} = \frac{1}{D_0} \mathcal{T} + \frac{1}{D_0} [2s \cdot \Psi - \Phi], \quad \mathcal{T} \equiv \int d^2 r \frac{e^{iq \cdot r}}{r^2}.$$

3. $\mathcal{T}$ in momentum space — it is the one-loop transverse bubble

$1/r^2$ is the square of the Weizsäcker–Williams kernel, whose transform is

$$\int d^2 r e^{i\ell \cdot r} \frac{r^i}{r^2} = \frac{2\pi i \ell^i}{\ell^2}.$$

By the convolution theorem, with $\frac{1}{r^2} = \frac{r^i}{r^2} \frac{r^i}{r^2}$,

$$\mathcal{T} = \int d^2 r e^{iq \cdot r} \frac{1}{r^2} = \int \frac{d^2 \ell}{(2\pi)^2} \frac{2\pi i \ell^i}{\ell^2} \frac{2\pi i (q - \ell)^i}{(q - \ell)^2} = (2\pi)^2 \int \frac{d^2 \ell}{(2\pi)^2} \frac{\ell \cdot (q - \ell)}{\ell^2 (q - \ell)^2}.$$

The divergence is now where $\overline{\text{MS}}$ expects it: at large ℓ the integrand tends to $1/\ell^2$, logarithmically divergent. There is no infrared problem — at $\ell \rightarrow 0$ the numerator $\ell \cdot (\ell - q) \rightarrow -\ell \cdot q$ is odd and averages to zero, and likewise at $\ell \rightarrow q$.

4. The loop in $d = 2 - 2\epsilon$

Continue the loop measure in the standard way (the two $2\pi i \ell^i/\ell^2$ are vertex factors and stay in two dimensions):

$$\mathcal{T}_\epsilon = (2\pi)^2 \mu^{2\epsilon} \int \frac{d^d \ell}{(2\pi)^d} \frac{\ell \cdot (q - \ell)}{\ell^2 (\ell - q)^2}.$$

Reduce with $\ell \cdot (\ell - q) = \ell^2 - \ell \cdot q$. The denominator is symmetric under $\ell \rightarrow q - \ell$, so $\int \frac{\ell \cdot q}{\ell^2(\ell - q)^2} = \frac{q^2}{2} \int \frac{1}{\ell^2(\ell - q)^2}$, while $\int \frac{d^d \ell}{(\ell - q)^2}$ is scaleless and vanishes:

$$\mathcal{T}_\epsilon = -\frac{q^2}{2} (2\pi)^2 B, \quad B \equiv \mu^{2\epsilon} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{\ell^2(\ell - q)^2} = \frac{\mu^{2\epsilon}}{(4\pi)^{d/2}} \frac{\Gamma(2 - \frac{d}{2})\Gamma(\frac{d}{2} - 1)^2}{\Gamma(d - 2)} (q^2)^{\frac{d}{2} - 2}.$$

At $d = 2 - 2\epsilon$, using $\frac{\Gamma(-\epsilon)^2}{\Gamma(-2\epsilon)} = -\frac{2}{\epsilon} \frac{\Gamma(1 - \epsilon)^2}{\Gamma(1 - 2\epsilon)}$,

$$B = \frac{1}{4\pi q^2} \left(\frac{4\pi\mu^2}{q^2} \right)^\epsilon \left(-\frac{2}{\epsilon} \right) c_\Gamma, \quad c_\Gamma = \frac{\Gamma(1 + \epsilon)\Gamma(1 - \epsilon)^2}{\Gamma(1 - 2\epsilon)} = 1 - \gamma_E \epsilon + O(\epsilon^2),$$

$$\mathcal{T}_\epsilon = \frac{\pi}{\epsilon} c_\Gamma \left(\frac{4\pi\mu^2}{q^2} \right)^\epsilon = \pi \left[\frac{1}{\epsilon} - \gamma_E + \log \frac{4\pi\mu^2}{q^2} \right] + O(\epsilon) = \pi \left[\frac{1}{\bar{\epsilon}} + \log \frac{\bar{\mu}^2}{q^2} \right].$$

5. $\overline{\text{MS}}$

Subtract $\frac{1}{\bar{\epsilon}} = \frac{1}{\epsilon} - \gamma_E + \log 4\pi$:

$$\mathcal{T}|_{\overline{\text{MS}}} = \pi \log \frac{\mu^2}{q^2}, \quad \text{equivalently} \quad \mathcal{T}_\epsilon = \pi \left[\frac{1}{\epsilon} + \log \frac{\bar{\mu}^2}{q^2} \right], \quad \bar{\mu}^2 = 4\pi\mu^2 e^{-\gamma_E}.$$

Why the pole is $+\frac{1}{\epsilon}$ here and $-\frac{1}{\epsilon}$ if you continue the coordinate integral. Continuing $\int d^{2-2\epsilon} r e^{iq \cdot r} / r^2$ directly gives $\pi \Gamma(-\epsilon) \left(\frac{q^2}{4\pi\mu^2} \right)^\epsilon = -\pi \left[\frac{1}{\epsilon} + \gamma_E + \log \frac{q^2}{4\pi\mu^2} \right]$. The two differ by *exactly* $2\pi/\epsilon$ — a pure pole, no finite part (checked numerically, §10D). The reason is the scaleless integral dropped in §4: $\int \frac{d^d \ell}{(\ell - q)^2} = 0 = \frac{1}{\epsilon_{\text{UV}}} - \frac{1}{\epsilon_{\text{IR}}}$, which trades the genuine large- ℓ ultraviolet pole for the bubble's small- ℓ infrared one. **The $\overline{\text{MS}}$ finite part — the only thing that enters the answer below — is identical either way: $\pi \log(\bar{\mu}^2/q^2)$.** The momentum-space form is the one that pairs with the standard $\overline{\text{MS}}$ counterterms, so it is used here.

6. The remainder $\mathcal{R}$, in full

The second piece of §2 needs no regulator: the bracket vanishes linearly at $r = 0$. Using $\frac{1}{D} - \frac{1}{D_0} = \frac{2r \cdot s - r^2}{D D_0}$ and $\frac{r^2}{r^2 D} = \frac{1}{D}$,

$$\mathcal{R} \equiv \int d^2 r e^{iq \cdot r} \frac{1}{r^2} \left[\frac{1}{D} - \frac{1}{D_0} \right] = \frac{1}{D_0} \int d^2 r e^{iq \cdot r} \frac{2r \cdot s - r^2}{r^2 D} = \frac{2s \cdot \Psi - \Phi}{D_0},$$

$$\Psi^i \equiv \int d^2 r e^{iq \cdot r} \frac{r^i}{r^2 D}, \quad \Phi \equiv \int d^2 r e^{iq \cdot r} \frac{1}{D}.$$

Φ in closed form. Shift $r = s + u$: $\Phi = e^{iq \cdot s} \int d^2 u \frac{e^{iq \cdot u}}{u^2 + M^2}$, so

$$\Phi = 2\pi e^{iq \cdot s} K_0(M|q|).$$

Ψ^i . Feynman-parametrise $\frac{1}{r^2 D}$ with $A = r^2$, $B = D$: $\alpha A + \bar{\alpha} B = (r - b)^2 + \Delta$, $b = \bar{\alpha} s$, $\Delta = \bar{\alpha}(\alpha s^2 + M^2)$ (no m — this is finite). Shift $r = b + u$, so $e^{iq \cdot r} = e^{i\bar{\alpha} q \cdot s} e^{iq \cdot u}$ and $r^i = b^i + u^i$:

$$\Psi^i = \int_0^1 d\alpha e^{i\bar{\alpha} q \cdot s} [b^i U_2 + V_2^i], \quad U_2 = \frac{\pi|q|}{\sqrt{\Delta}} K_1(\lambda), \quad V_2^i = i\pi q^i K_0(\lambda), \quad \lambda = |q|\sqrt{\Delta},$$

the two transforms following from $\int \frac{d^2 u e^{iq \cdot u}}{(u^2 + \Delta)^2} = -\partial_\Delta [2\pi K_0(|q|\sqrt{\Delta})]$ and $V_2^i = -i\partial U_2 / \partial q_i$. Contracting with s :

$$s \cdot \Psi = \pi \int_0^1 d\alpha e^{i\bar{\alpha} q \cdot s} \left[\bar{\alpha} s^2 \frac{|q|}{\sqrt{\Delta}} K_1(\lambda) + i(q \cdot s) K_0(\lambda) \right], \quad \Delta = \bar{\alpha}(\alpha s^2 + M^2).$$

Equivalently, as one convergent parameter integral (Feynman parameter $t = \bar{\alpha}$, the $t \rightarrow 0$ subtraction removing what is already in $\mathcal{T}$),

$$\mathcal{R} = \frac{\pi}{D_0} \left[\hat{I} + \log \frac{e^{2\gamma_E} q^2 D_0}{4} \right], \quad \hat{I} = D_0 \int_0^1 dt \left[e^{itq \cdot s} \frac{|q| K_1(|q| \sqrt{\Delta_0})}{\sqrt{\Delta_0}} - \frac{1}{t D_0} \right],$$

$$\Delta_0 = t[(1-t)s^2 + M^2].$$

Checked against direct two-dimensional quadrature to 10^{-9} , and the two forms against each other to 10^{-15} (§10E).

7. Adding the two pieces — the q^2 cancels

$$\mathcal{G}\Big|_{\overline{\text{MS}}} = \frac{\mathcal{T}_{\overline{\text{MS}}}}{D_0} + \mathcal{R} = \frac{\pi}{D_0} \left[\underbrace{\log \frac{\mu^2}{q^2}}_{\text{from } \mathcal{T}} + \underbrace{\log \frac{e^{2\gamma_E} q^2 D_0}{4}}_{\text{from } \mathcal{R}} + \hat{I} \right] = \frac{\pi}{D_0} \left[\log \frac{e^{2\gamma_E} \mu^2 D_0}{4} + \hat{I} \right].$$

The q^2 drops out between the two. This matters: with $D_0 = s^2/\bar{\xi}$ the logarithm is of the *dipole size*, not of the produced-gluon momentum,

$$\mathcal{G}\Big|_{\overline{\text{MS}}} = \frac{\pi}{D_0} \left[\log \frac{\mu^2 (y-x)^2}{\bar{\xi}} + 2(\gamma_E - \log 2) + \hat{I}(\xi) \right],$$

and the $\overline{\text{MS}}$ scheme constant appears as $2(\gamma_E - \log 2)$. (Verified to 10^{-15} , §10F. This supersedes the reading of the earlier draft, where only $\mathcal{T}$ was kept and the log appeared to carry k^2 .)

8. Back into the row, and the p^+ integral

Since $\frac{1}{\bar{\xi}} \cdot \frac{1}{D_0} = \frac{1}{s^2}$, the $\bar{\xi}$ cancels:

$$\frac{1}{\bar{\xi}} \frac{s \cdot P}{P^2} C_{\text{UV}}(\xi) \mathcal{G} = \pi C_{\text{UV}}(\xi) \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} \left[\log \frac{\mu^2 (y-x)^2}{\bar{\xi}} + 2(\gamma_E - \log 2) + \hat{I}(\xi) \right].$$

The three ξ moments needed are

$$\int_{\lambda}^{1-\lambda} d\xi C_{\text{UV}} = 2\ell_k - \frac{11}{6}, \quad \int_{\lambda}^{1-\lambda} d\xi C_{\text{UV}} \log \frac{1}{\bar{\xi}} = \frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36},$$

$$\int_{\lambda}^{1-\lambda} d\xi C_{\text{UV}}(\xi) \hat{I}(\xi) = e^{ik \cdot (y-x)} \frac{\ell_k^2}{2} + c_0 \ell_k + c_1,$$

the last fixed by the endpoints: $\hat{I} \rightarrow e^{ik \cdot (y-x)} \log \frac{1}{\bar{\xi}} + c_0$ as $\xi \rightarrow 0$, with

$$c_0 = s^2 \int_0^1 dt \left[e^{itk \cdot s} \frac{k K_1(k s \sqrt{t(1-t)})}{s \sqrt{t(1-t)}} - \frac{1}{t s^2} - \frac{e^{ik \cdot s}}{s^2(1-t)} \right],$$

while $\hat{I} \rightarrow 0$ as $\xi \rightarrow 1$, so that endpoint gives no rapidity logarithm from $\hat{I}$.

9. Final result

Prefactor $\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4 k^+} \times \frac{\pi}{2\pi} = \frac{\alpha_s^2}{8\pi^5 k^+}$:

$$\frac{d^3 N_2^{\text{NLO}}}{d^3 k} = \frac{\alpha_s^2(\mu) N_c}{8\pi^5 k^+} \int_{x,y,x',w'} e^{-ik \cdot w'} e^{ik \cdot x} \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} [\mathcal{P} + \mathcal{F}] \\ \times [U^{ab'}(x') - U^{ab'}(w')] [U^{af}(y) - U^{af}(x)] \rho^{b'}(x') \rho^f(y),$$

$$\mathcal{P} = \frac{1}{\hat{\epsilon}} \left(2\ell_k - \frac{11}{6} \right) \xrightarrow{\overline{\text{MS}}} 0$$

$$\mathcal{F} = \left(2\ell_k - \frac{11}{6} \right) \left[\log(\mu^2(y-x)^2) + 2(\gamma_E - \log 2) \right] + \frac{1 + e^{ik \cdot (y-x)}}{2} \ell_k^2 + \frac{\pi^2}{6} - \frac{67}{36} + c_0 \ell_k + c_1$$

The scheme constant is the paper's. $(2\ell_k - \frac{11}{6}) \times 2N_c = 4N_c \ell_k - \frac{11N_c}{3} = 4N_c \ell_k - b$ at $N_f = 0$, so the constant riding on it enters as

$$2b(\gamma_E - \log 2),$$

which is exactly the term in K_{JSJ} , Eq. (2.64) of arXiv:1610.03453 — the one their footnote 2 records as having been missing from their earlier results. It comes out here only because $\mathcal{R}$ was kept: it is the $\log \frac{e^{2\gamma_E}}{4}$ left behind when the q^2 cancels in §7.

Two further readings. The ℓ_k^2 coefficient is $\frac{1}{2}(1 + e^{ik \cdot (y-x)})$, one half from each endpoint of the ξ integral ($\xi \rightarrow 1$ through $\log \frac{1}{1-\xi}$ against the $\frac{\xi}{1-\xi}$ pole, $\xi \rightarrow 0$ through $\hat{I}$'s $z \rightarrow y$ logarithm against the $\frac{1-\xi}{\xi}$ pole); it is regulator-independent, and agrees with the mass-regulator computation. And $\mathcal{P} \times 2N_c = \frac{1}{\epsilon}(4N_c \ell_k - b)$ is the coupling renormalisation, cancelling against the $-2/\epsilon$ of the self-energy counterterm.

10. Checks

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A. the d-dimensional transform, formula checked where the answer is known (d=3, a=1)
pi^(d/2) 2^(d-2a) Gamma((d-2a)/2)/Gamma(a) = 19.739208802179
known result int d^3x e^(iqx)/x^2 = 2 pi^2/|q| , 2 pi^2 = 19.739208802179

B. the epsilon expansion: pi Gamma(-eps) (q^2/(4 pi mu^2))^eps + pi/eps -> pi log(mubar^2/q^2)
q^2 = 2.7, mu^2 = 1.3, mubar^2 = 4 pi mu^2 e^-gamma = 9.17216034208
eps= 1e-3: Gamma-form + pi/eps = 3.8369896778214 pi log(mubar^2/q^2) = 3.8419198605062
eps= 1e-4: Gamma-form + pi/eps = 3.8414265852134 pi log(mubar^2/q^2) = 3.8419198605062
eps= 1e-5: Gamma-form + pi/eps = 3.8418705304043 pi log(mubar^2/q^2) = 3.8419198605062
eps= 1e-6: Gamma-form + pi/eps = 3.8419149274703 pi log(mubar^2/q^2) = 3.8419198605062
eps= 1e-7: Gamma-form + pi/eps = 3.8419193672023 pi log(mubar^2/q^2) = 3.8419198605062

C. the p+ moments (lambda = 1e-9, l_k = log(1/lambda))
int C_UV = 39.613198342559 2 l_k - 11/6 = 39.613198340559
int C_UV log(1/xib^2) = 429.02139290167 l_k^2 + pi^2/3 - 67/18 = 429.02139286022
int C_UV log(xi/xib) = -7.6498852e-22 (odd under xi <-> xibar, vanishes)
int C_UV log(xi xib) = -429.02139290167 [paper (C.10)]; sum with the row above: -7.6498848e-22

D. MS-bar in momentum space: the 1/r^2 term as the one-loop transverse bubble
Gamma(-e)^2/Gamma(-2e) = -(2/e) Gamma(1-e)^2/Gamma(1-2e) :
e= 0.3: -5.06410115646405 -5.06410115646405 diff -7.889e-31
e= 0.05: -39.82284747944915 -39.82284747944915 diff 6.311e-30
e= -0.02: 99.93608767676722 99.93608767676722 diff -1.262e-29
the two routes (q^2=2.7, mu^2=1.3):
e=0.001: momentum finite part 3.8416741502922 coordinate finite part 3.8369896778214
pi log(mubar^2/q^2) = 3.8419198605062 difference of routes 6283.1899916521 vs 2 pi/e =
6283.1853071796
e=0.00001: momentum finite part 3.8419175126647 coordinate finite part 3.8418705304043
pi log(mubar^2/q^2) = 3.8419198605062 difference of routes 628318.53076494 vs 2 pi/e =
628318.53071796
e=0.0000001: momentum finite part 3.8419198370387 coordinate finite part 3.8419193672023
pi log(mubar^2/q^2) = 3.8419198605062 difference of routes 62831853.071796 vs 2 pi/e =
62831853.071796
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zint/src/run_dimreg.py — the numbers above.